THS WindBreakers

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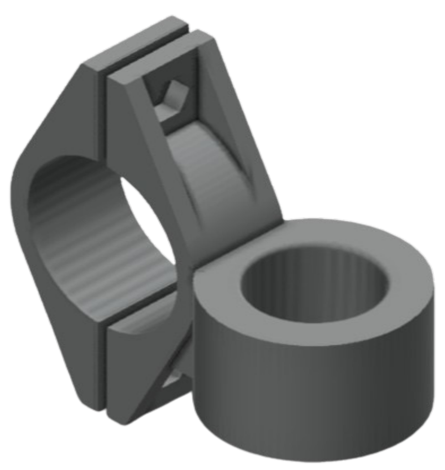
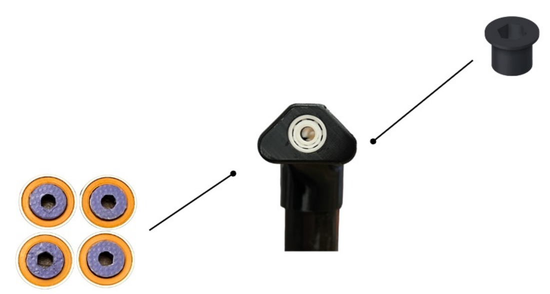
Gearbox

Gears
•Using stock gears at a 32:1 gear ratio
•Have plans to design our own gears and hub
•Can easily swap gears for differing speeds
•Created gear caps to hold gear rods and protect bearings from debris
•Oriented in a fashion for a horizontal axis wind turbine

Motor
•Used the KidWind stock DC motor
•Great for high torque applications
•Created an easily adjustable mount for its placement



Bearings
•Started with steel but upgraded to ceramic bearings for:
•Longer spin time
•Less friction
•Better acoustics

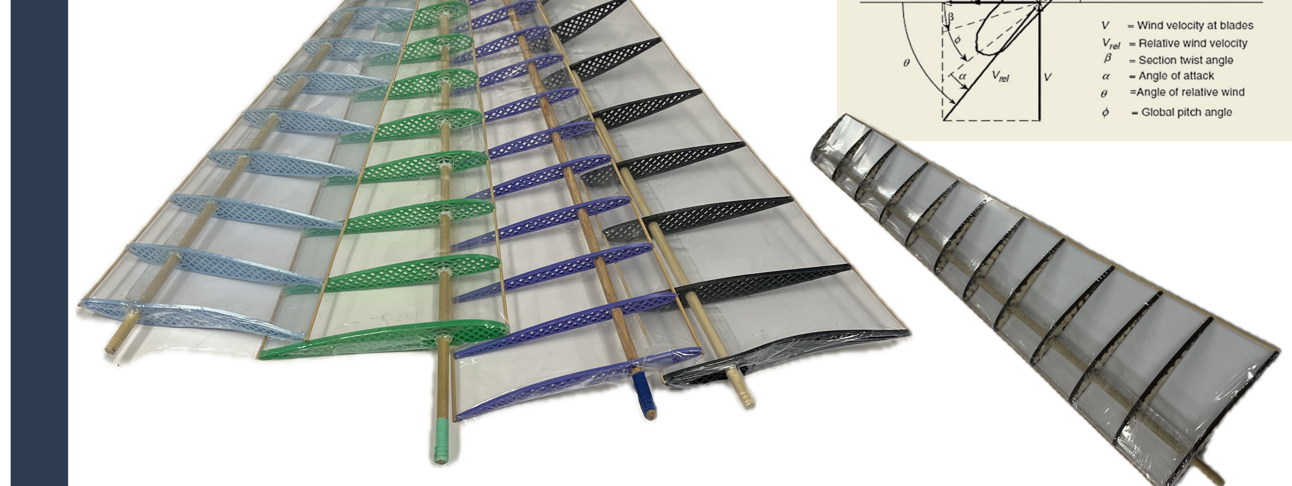


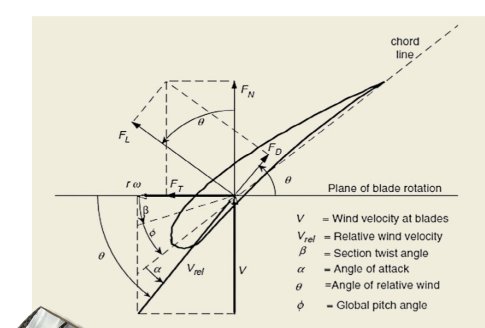
Materials
•Utilized metal pipe and metal floor flange to mount the gearbox onto a wooden board
•Wooden board thickness is 1 inch, and our metal pipe length is 23 inches, putting us in the center of the wind tunnel at 24 inches, maximizing our energy potential

Iterations
•Redesigned each part for better strength and adjustability
•Created hex slots to secure parts with screws
•Made a two-piece clamping design for better holding

Blades

- First used window insulation to wrap blades but this material resulted in dips and extra weight
  - Our current wrapping material is a thin plastic sheet typically used to wrap gift baskets
  - Offers greater strength for its weight and stretches to prevent dipping
- Twisted our blades 13° for most optimized twist for the best angle of attack at varied distances and also reduces stress on the blade
- First started with a five-airfoil design, but dips were created when heat shrinking the plastic, reducing the effective blade shape so increased to 12
- Used wooden dowel rod as the main structure and set airfoils onto it accordingly
- Used bamboo edges to complete the blades





Airfoils

First Design
•First modeled NACA 23012 12%
•It is a typical model RC Plane airfoil
•Low Reynolds number airfoil (~23,000)

Updated Airfoil
•Redesigned airfoil shape specifically created for low-speed applications (~5m/s)
•Our new S1210 airfoils have greater camber resulting in higher lift capabilities
•We designed a flat back to maintain structural integrity and make it easier to produce
•Numbered airfoils with laser

3D Printed Airfoils
•We first 3D printed 5 airfoils, increased it to 9, and finally 12 to best maintain blade profile
•Thickness of 3mm to maximize strength to weight ratio
•Tolerances of +.15mm were added to the holes for proper fit
•PLA was used due to its high strength to weight ratio and ease of use
•Implemented grid infill structure to maintain high strength but reduce excess weight

Laser Cut Airfoils
•Switched to basswood laser cut airfoils for faster production and easier iterating
•Maintained the 3mm thickness but was much stronger than 3D printed design
•Allowed us to make smaller, more specific airfoils with same strength but at the cost of more weight

Blade Construction

1. Mark Foil Positions:
•Purchased Oak dowel rod
•Cut it to 20 inches
•Marked the airfoil rib section for 12 airfoils

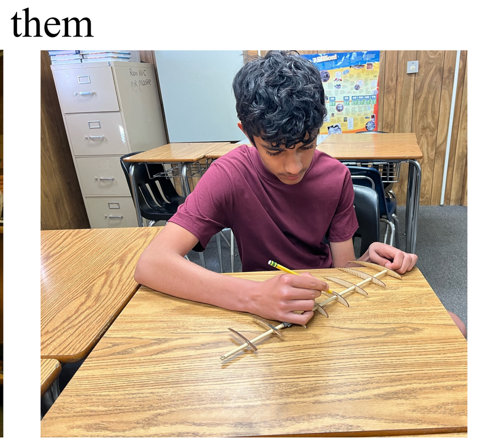
2. Glue Foils:
•Added the airfoils to the marked ribs
•Twisted the airfoils (~1° for each)
•Applied super glue to secure

•3. Glue Edges:
•Inserted thin bamboo rods along leading and trailing edges
•Super glued the rods onto airfoils

4. Wrap Blades:
•Laid double-sided tape along the airfoils for wrap to stick
•Added layer of tape in between blades to minimize dips
•Rolled the blades to fully encapsulate with no wrinkles
•Use hair dryer to shrink the wrap and conform the shape of the airfoils

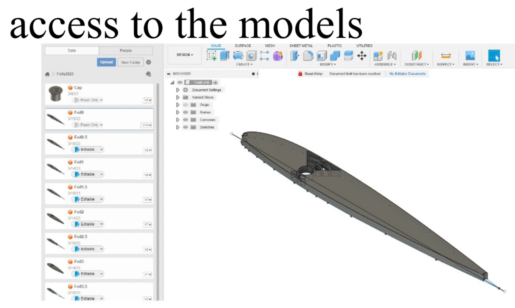
Iterations

- The heat shrink created massive dips in the blade between the airfoils so we:
  - Added more airfoils and a thin bamboo strip, but this was too heavy
  - Instead, we added tape on the interior and then wrapped shrink wrap around it
- Plastic wrap was originally too weak and ripped easily -> Then we used window insulation, which yielded around the same results -> Finally, we wrapped the blades with a thin plastic sheet typically used to wrap gift baskets, which we found was both durable and flexible, offering much more strength than previous options.
- Tested various edge materials – some were too flimsy, some were too big, and others were too weak
  - We finally decided on bamboo for the edges, as it has the three times the strength-to-weight ratio compared to steel (28000 lb/sq. inch tensile strength)
  - Bamboo sticks are 2mm in diameter
  - These types of bamboo sticks are rare, so a team member had to go to Japan to fetch them



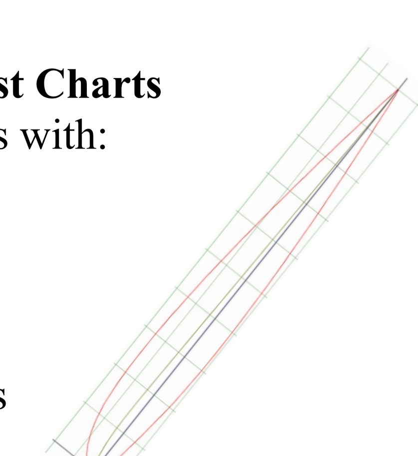
Programs

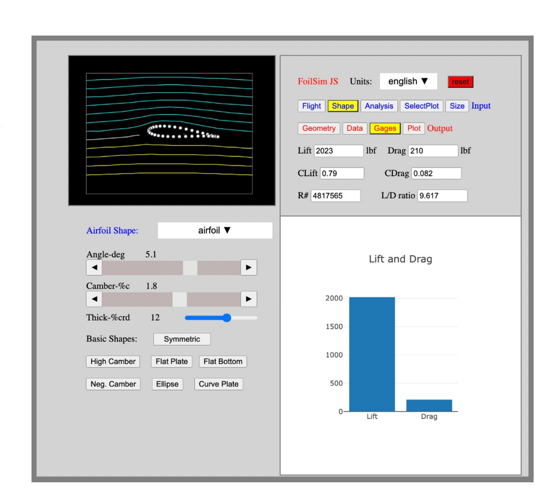
Fusion 360
•Timeline editing allows for easy modification remotely
•Document sharing allows everyone access to the models
•Easy file exporting to for printing

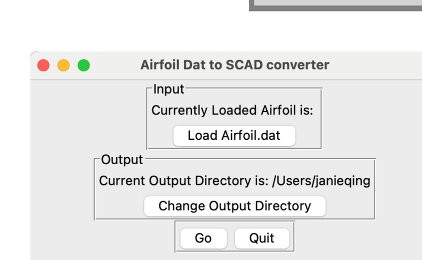


Python Foil Maker
•Input airfoil .dat file
•Outputs SCAD file
•SCAD file is 3D foil w/ holes for edges and rod
•SCAD file can be exported as STL to be printed

Optimal Twist Charts
Created charts with:
•Pitch Setting
•Blade Twist
•Root Chord
•Tip Chord
•Blade Radius
•Wind speed
•RPM
Gives Angle of Attack for each section of the blade







Airfoil Simulation
•Used NASA Airfoil Simulation program
•Changed the angle degree based on the maximum lift vs minimum drag ratio.

Testing

Custom Wind Tunnel
•Built a custom wind tunnel at Tabb High for precise testing purposes
•Used Vernier Graphical Analysis software and a multimeter to analyze and measure our turbine's electrical output
•Wind speed of our wind tunnel is 2.7 m/s using a single fan

Various Tests Conducted
•Blade pitch
•Distance to the fan
•Direction of gearbox
•High torque generator and regular generator
•Number of blades
•Blade Twist

Documentation
•The bottom table illustrates the various variables changes tested during
testing with our wind tunnel
- All data were recorded in our KidWind engineering notebook